UGSF 1st Month Report (1st Sept – 29th Sept, 2025)

Project : TrackMate Pro

Description : A Next.js and MongoDB-powered platform designed to streamline student project workflows for SGP

Team Members : Kartik Guleria – 23DIT015

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DATE	Module Worked upon	Hours worked on that module	Brief about what have been done
1-9-2025	Project Planning & System Architecture	2	Analyzed requirements and designed overall system architecture for TrackMate Pro student portal
2-9-2025	Backend API Architecture Design	3	Created Next.js API routes structure and designed RESTful endpoints for authentication and user management
3-9-2025	Authentication System Implementation	3	Implemented NextAuth.js configuration and college email validation system (YYDEPTXXX@charusat.edu.in)
4-9-2025	User Management API Development	2	Created user registration API endpoints and implemented role-based route protection middleware
5-9-2025	Database Models & Schema Design	1	Created comprehensive User, Subject, Assignment and Submission models with relationships and database indexes
6-9-2025 & 7-9-2025 → Weekends			
8-9-2025 to 11-9-2025 → Internal Examination			
12-9-2025	Assignment Management API	2	Developed assignment creation endpoints and implemented file upload functionality with Cloudinary integration
13-9-2025 & 14-9-2025 → Weekends			
15-9-2025	Project Group Management System	1	Created ProjectGroup model and implemented group formation with HOD approval workflow APIs
16-9-2025	Real-time Notification System	1	Integrated Socket.io for real-time notifications and implemented notification API endpoints for different user roles

17-9-2025	Admin Panel APIs	3	Developed admin dashboard APIs for user management and implemented counsellor assignment functionality
18-9-2025	Grade Management & Analytics	2	Created grading system with feedback mechanisms and implemented academic analytics with PDF report generation
19-9-2025	Security & Performance Optimization	1.5	Implemented comprehensive input validation and optimized database queries with caching mechanisms
20-9-2025 & 21-9-2025 → Weekends			
22-9-2025	API Testing & Documentation	2	Created comprehensive API documentation and implemented unit tests for critical endpoints

Total Hours Worked this month – 23.5 hours